

UNCERTAINTY-CALIBRATED qPCR MARKER PANELS WITH EMPIRICAL DETECTION AND QUANTIFICATION LIMITS

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ABSTRACT

Quantifying rare cell populations from targeted molecular measurements requires marker panels that are not only discriminative, but also explicit about when their abundance estimates can be trusted. We develop an uncertainty-aware framework for qPCR-style marker-panel selection and cell-fraction estimation from single-cell RNA-sequencing data. The framework separates discovery, panel selection, uncertainty calibration, and held-out evaluation; combines discovery-only marker ranking with cross-validated fraction modeling; and constructs split-conformal and residual-bootstrap prediction intervals from independently generated calibration pseudobulks. To assess robustness, we simulate finite-cell sampling and fixed assay nuisances, including global scaling, gene-wise noise, dropout, ambient RNA, and altered background composition, while evaluating explicit target-abundance regimes from trace to high abundance. Across the completed uncertainty-evaluation configurations, multi-gene panels generally improved quantitative error over the best single-marker baseline, with panel mean absolute error ranging from 0.059 to 0.064 compared with 0.0622 to 0.0724 for single markers, while panel-minus-single-marker AUC differences remained modest. Empirical conformal coverage was below nominal levels, ranging from 56.2% to 70.5% for nominal 80% intervals and from 73.8% to 88.8% for nominal 95% intervals, demonstrating that point-estimation performance does not guarantee reliable uncertainty calibration under distributional and assay variation. These results support reporting empirical detection and quantification limits and treating uncertainty calibration as a separate experimental objective rather than inferring measurement reliability from discrimination or average error alone.

1 INTRODUCTION

Measuring the abundance of a rare cell population from a small set of molecular markers is a central problem in targeted transcriptomic and qPCR-style assays. A useful marker panel should distinguish the target from background populations, but discrimination alone is insufficient when the scientific goal is to estimate cell fractions. In that setting, an assay must also indicate when a target is below detection, when it is detected but cannot be quantified reliably, and when its estimated abundance is sufficiently trustworthy for downstream use. Explicit empirical limits of detection and quantification are therefore as important as average predictive accuracy.

Several factors make this problem difficult. Marker expression can vary across biological populations, while finite-cell sampling, global RNA variation, gene-specific assay noise, dropout, ambient RNA, and changes in background composition can distort the measured signal. Rare targets are especially challenging because small absolute errors can substantially affect decisions near the detection boundary. Moreover, marker discovery, panel selection, model fitting, uncertainty calibration, and held-out evaluation can leak information into one another if they are not separated. Consequently, a panel may have strong held-out discrimination or low average fraction error while still producing prediction intervals that fail to achieve their nominal coverage.

We address these challenges with an uncertainty-aware framework for qPCR-style marker-panel selection and cell-fraction estimation from single-cell RNA-sequencing data. The framework uses

cluster-stratified discovery and holdout cell splits, performs marker ranking only on discovery cells, and generates independent pseudobulk sets for panel selection, uncertainty calibration, and held-out evaluation. Candidate panels are evaluated using cross-validated fraction error together with calibration-set residual behavior, rare-fraction error, prediction-interval width, and a small panel-size penalty. After fitting a linear fraction model on the panel-selection pseudobulks, we construct clipped split-conformal intervals from calibration residuals and evaluate an optional residual-bootstrap alternative. The evaluation fixes the nuisance-parameter values governing assay and background perturbations before held-out testing, while pseudobulk sampling and random perturbation draws are applied independently to the generated profiles; target fractions are sampled from explicit abundance regimes, including trace and very-low abundance.

Our contributions are:

- We formulate marker-panel evaluation as a joint problem of detection, quantitative fraction estimation, and uncertainty calibration rather than as a discrimination problem alone.
- We provide a leakage-controlled workflow that separates discovery-only marker ranking, panel selection, calibration, and held-out evaluation.
- We introduce uncertainty-aware forward selection that combines cross-validated fraction error with rare-fraction error, residual-based interval width, and panel complexity.
- We evaluate split-conformal and residual-bootstrap intervals under finite-cell sampling and fixed assay nuisance perturbations across predefined target-abundance regimes.
- We use completed held-out experiments and generated diagnostics to examine the trade-offs among panel-versus-single-marker performance, point-estimation error, interval coverage, and interval width without treating nominal coverage as an experimentally established fact.

The completed held-out experiments show that multi-gene panels generally improve point-estimation error over best single-marker baselines, while their detection gains remain modest. In contrast, split-conformal intervals fall substantially below nominal coverage: across the completed configurations, empirical coverage ranges from 56.2% to 70.5% for nominal 80% intervals and from 73.8% to 88.8% for nominal 95% intervals. These results show that useful point estimates and strong detection ranking do not establish calibrated uncertainty, motivating transparent empirical assay limits rather than unsupported confidence in point estimates alone.

2 RELATED WORK

3 BACKGROUND

3.1 PROBLEM SETTING

We consider a heterogeneous sample containing cells from a finite set of populations $\mathcal{C} = \{c_1, \dots, c_K\}$. For each cell, a transcriptomic measurement vector $\mathbf{x} \in \mathbb{R}_{\geq 0}^G$ is observed over G genes, together with a population label. For a target population $c \in \mathcal{C}$, the objective is to estimate its fraction $f \in [0, 1]$ in a bulk-like sample from a small marker panel. The same estimate can also be used for a binary detection decision, for example by comparing the true or predicted fraction with a prespecified detection threshold. Thus, the task combines continuous fraction estimation with target-population discrimination.

A pseudobulk sample is constructed by drawing n cells, with n fixed by the assay design, and averaging their expression vectors:

$$\mathbf{b} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i.$$

If n_c of these cells belong to the target population, the realized target fraction is $f = n_c/n$. The remaining cells are sampled from the non-target populations according to a background mixture. This construction assumes that the observed single-cell profiles provide representative draws from the populations of interest and that averaging is an adequate proxy for the targeted molecular measurement. It also makes finite-cell sampling part of the measurement problem: two samples with the same

population proportions can produce different molecular summaries because they contain different cells.

For a candidate marker panel $M_c \subseteq \{1, \dots, G\}$ and a reference set $R \subseteq \{1, \dots, G\}$, we summarize a bulk sample using a qPCR-style delta-expression proxy. Specifically, the panel score is

$$s(\mathbf{b}; M_c, R) = \frac{1}{|M_c|} \sum_{g \in M_c} b_g - \frac{1}{|R|} \sum_{g \in R} b_g.$$

The reference genes are selected from genes that are expressed across the discovery populations and have relatively stable between-population expression, while candidate markers are selected separately for each target. Because the input data contain expression values rather than physical cycle-threshold measurements, this score should be interpreted as a normalization proxy rather than as a calibrated ΔC_t measurement.

Given training pseudobulks $(\mathbf{b}_j, f_j)$, we fit a one-dimensional linear regression,

$$f_j = \beta_0 + \beta_1 s(\mathbf{b}_j; M_c, R) + \varepsilon_j,$$

and clip its predictions to the physically meaningful interval $[0, 1]$. The model is deliberately simple: its purpose is to provide an interpretable mapping from a marker score to a cell fraction while allowing panel construction and uncertainty assessment to be evaluated independently. The resulting point estimate is used both for quantitative error metrics and for ranking samples in detection analyses.

3.2 MARKER DISCOVERY AND PANEL EVALUATION

Marker candidates are obtained by ranking genes for each population using only discovery cells. In the completed pipeline, this ranking uses the Wilcoxon procedure implemented in the single-cell analysis toolkit. The cells are split in a cluster-stratified manner before marker ranking, so held-out cells do not influence candidate discovery, reference-gene selection, or model fitting. This separation is important because marker selection performed on the same observations used for evaluation can make apparent discrimination or quantitative accuracy optimistic.

For each target, the best single marker is selected from the candidate list by cross-validated fraction MAE on the panel-selection pseudobulks. A multi-gene panel is then built by greedy forward selection, beginning with the best single marker and adding a candidate only when the cross-validated MAE improves by at least a fixed minimum gain, up to the allowed panel size. This provides a direct baseline comparison: the single marker tests whether one gene is sufficient, whereas the panel tests whether aggregating several target-associated signals improves fraction estimation or detection.

An uncertainty-aware panel-selection objective augments cross-validated MAE with calibration-set rare-fraction MAE, conformal residual widths at 80% and 95% coverage levels, and a small penalty for panel size. Candidate panels are scored using the panel-selection and calibration pseudobulks only. The held-out pseudobulks are reserved for final evaluation and therefore cannot determine the selected panel or the interval widths. This distinction separates optimizing a useful assay summary from validating how well that summary transfers to new samples.

3.3 PREDICTION UNCERTAINTY AND ASSAY LIMITS

After fitting the fraction model on panel-selection pseudobulks, let $\hat{f}_i^{\text{cal}}$ and f_i^{cal} denote calibration predictions and observed simulated fractions. We compute absolute calibration residuals

$$r_i = \left| \hat{f}_i^{\text{cal}} - f_i^{\text{cal}} \right|$$

and use their finite-sample empirical quantile q_α to form a nominal $(1 - \alpha)$ interval for a new prediction $\hat{f}$:

$$\left[\max(0, \hat{f} - q_\alpha), \min(1, \hat{f} + q_\alpha) \right].$$

The clipping reflects the bounded fraction domain. Nominal coverage is a calibration target, not an empirical guarantee under arbitrary shifts; transfer can fail when calibration and test pseudobulks differ in sampling, abundance composition, or assay nuisance conditions.

Three properties are consequently treated as distinct. Point accuracy concerns the distance between predicted and true fractions, summarized by metrics such as MAE and RMSE. Detection concerns ranking or threshold discrimination, summarized here by ROC AUC relative to a fixed detection threshold. Uncertainty calibration concerns whether intervals contain the true fraction at the stated rate, together with their width and behavior across abundance strata. A panel can therefore improve MAE or AUC without producing intervals with nominal coverage, and intervals can be wide without establishing useful quantitative resolution.

An empirical limit of detection is the lowest target abundance at which a predefined detection sensitivity can be achieved at a fixed specificity. An empirical limit of quantification is more stringent: it requires both adequate interval coverage and an error criterion such as an MAE threshold. In practice, these definitions induce three operational outcomes: below detection, detected but not reliably quantifiable, and quantifiable. The present study evaluates these concepts through explicit abundance regimes and held-out pseudobulks; the limits must be inferred from the corresponding completed evaluations rather than from nominal thresholds alone.

The measurement process is additionally perturbed by finite-cell sampling and fixed nuisance parameters representing global scaling, gene-wise noise, dropout, ambient RNA, and altered background composition. These perturbations are generated before held-out evaluation and are held fixed as an experimental realization, while new pseudobulks are sampled from discovery or held-out cells. This setup asks whether a panel and its uncertainty procedure remain informative when molecular summaries depart from idealized population averages, without treating robustness to one nuisance realization as universal assay validation.

4 METHOD

4.1 DATASET AND CELL-TYPE PARTITIONING

We use a single-cell PBMC expression dataset with cell-level cluster labels stored in the `leiden` annotation. Let $\mathbf{x}_i$ denote the expression vector for cell i and let $z_i \in \mathcal{C}$ denote its cluster. The cluster labels define the candidate target populations, the discovery and holdout stratification, and the background populations used during pseudobulk simulation. The resulting cell-type landscape is shown in fig. 1; the UMAP coordinates are used only for visualization and not as model features.

4.2 LEAKAGE-CONTROLLED DATA SPLITS

We first split cells within each cluster into discovery and holdout subsets using a fixed discovery fraction of 0.70. Each cluster is shuffled independently, with at least two cells retained in each subset. Marker ranking and reference-gene selection use only the discovery cells. Pseudobulks for model fitting and calibration are also generated from discovery cells, whereas held-out evaluation pseudobulks are generated exclusively from the holdout cells. This separation prevents held-out cell expression from affecting candidate discovery, panel selection, model fitting, or interval calibration.

For each target cluster c , we rank genes on the discovery cells using Scanpy’s Wilcoxon marker-ranking procedure and retain up to 30 candidate genes. The resulting target-specific candidate list is the only source of genes considered for that target’s panel. We also construct a reference-gene pool by ranking genes according to their relative between-cluster variability, requiring expression in at least 20% of cells and excluding all candidate marker genes. The first three genes in the resulting stable pool are used as reference genes for every target.

4.3 PSEUDOBULK SIMULATION AND ASSAY PERTURBATION

For each target cluster, we generate independent training, calibration, and test pseudobulk sets. A pseudobulk contains 500 sampled cells, with target abundance drawn from one of five fixed regimes: absent or trace (0–1%), very low (1–5%), low (5–10%), medium (10–50%), or high (50–90%). The target count is rounded to an integer and the realized fraction is recorded as $n_c/500$. The non-target cells are distributed across all other clusters using a Dirichlet draw followed by a multinomial draw, with concentration parameter 0.35 when the assay perturbation layer is active.

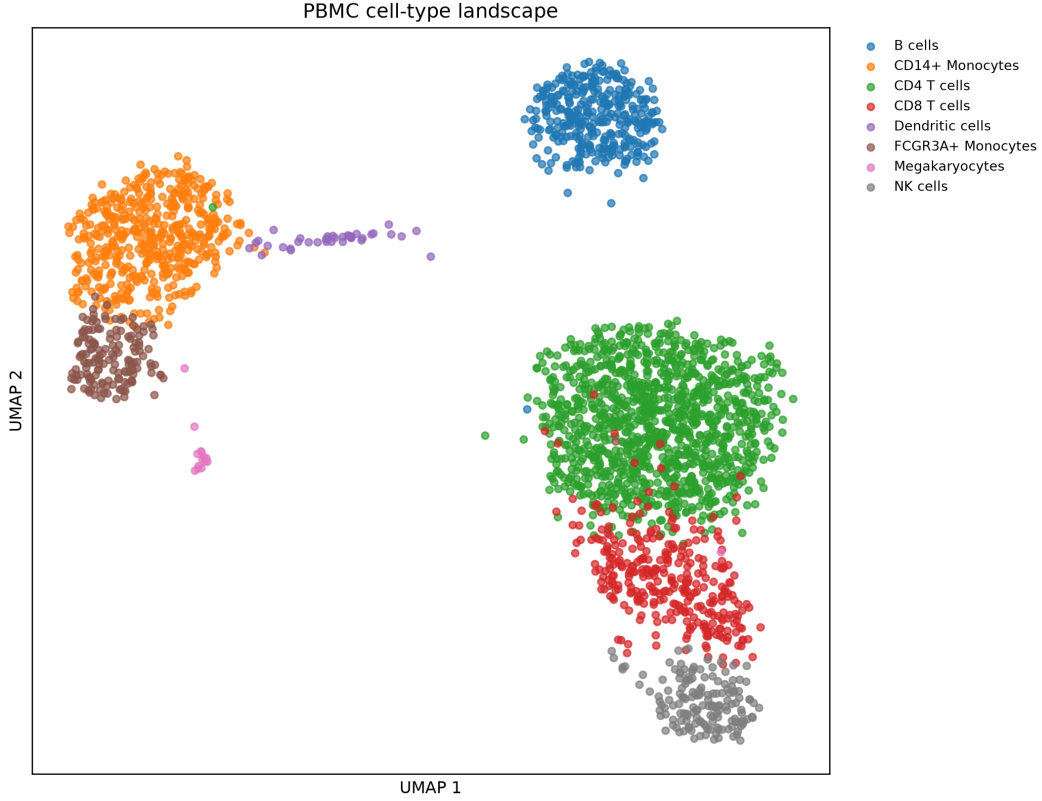


Figure 1: UMAP visualization of the single-cell PBMC dataset colored by cluster label. Each point is a cell; the embedding provides dataset context for discovery-only marker ranking and cluster-stratified splitting, rather than a quantitative abundance measurement.

Before generating the pseudobulk sets, we fix the nuisance-parameter values for the experiment: gene-specific multiplicative noise scales, a global scaling standard deviation of 0.08, a dropout probability of 0.02, an ambient-RNA fraction of 0.01, the discovery-cell mean expression profile as the ambient profile, and the altered background-mixture concentration. For each generated bulk profile, independent random draws determine the log-normal global scale, dropout mask, and other sampling perturbations; the profile is then multiplied by the global and gene-specific scales, has randomly selected entries set to zero according to the dropout probability, and is mixed with the ambient profile. Thus, parameter values are held fixed across training, calibration, and held-out generation, but the perturbation realizations and pseudobulk cell samples are independently drawn for each profile.

4.4 NORMALIZED MARKER SCORES AND FRACTION MODEL

For a panel M_c and reference set R , we compute the normalized score

$$s(\mathbf{b}; M_c, R) = \frac{1}{|M_c|} \sum_{g \in M_c} b_g - \frac{1}{|R|} \sum_{g \in R} b_g.$$

The score is a delta-expression proxy because the input matrix contains expression values rather than cycle-threshold measurements. For each target, we fit a linear regression from this one-dimensional score to the realized target fraction using the training pseudobulks. Predictions on calibration and test pseudobulks are clipped to $[0, 1]$.

We select the best single-marker baseline by evaluating each target candidate with five-fold shuffled cross-validated linear-regression predictions on the training pseudobulks and choosing the gene with the lowest clipped fraction MAE. Starting from this best single marker, nominal panels are

constructed by greedy forward selection. A remaining candidate is added only when it improves cross-validated MAE by at least 0.002, and selection stops at five markers or when no candidate satisfies the improvement criterion.

4.5 UNCERTAINTY-AWARE PANEL SELECTION

The final panel is selected with a separate greedy forward procedure that uses the training and calibration pseudobulks. For each trial panel, we fit the fraction model on the training set and evaluate predictions and residuals on the calibration set. The objective combines training-set five-fold cross-validated MAE, calibration-set MAE restricted to fractions at or below 0.50, the widths of the 80% and 95% residual intervals, and a panel-size penalty:

$$J(M_c) = \text{MAE}_{\text{CV}} + 0.25 \text{MAE}_{\text{rare}} + 0.10 (q_{0.80} + q_{0.95}) + 0.002 |M_c|.$$

At each step, the candidate producing the smallest objective is added if it strictly improves the best objective. The procedure considers at most five markers and falls back to the first valid candidate if no panel can be selected.

4.6 CONFORMAL AND BOOTSTRAP INTERVALS

For the selected panel, we calculate signed calibration residuals $\hat{f}_i^{\text{cal}} - f_i^{\text{cal}}$ and their absolute values. The same calibration pseudobulks contribute to uncertainty-aware panel selection through calibration-set error and residual widths, so the resulting intervals are empirical held-out diagnostics rather than intervals with the usual split-conformal validity interpretation for a panel fixed independently of calibration data. For each nominal coverage level $1 - \alpha \in \{0.80, 0.95\}$, we compute the finite-sample empirical quantile

$$q_{1-\alpha} = \text{Quantile}_{\lceil (m+1)(1-\alpha) \rceil} \left(\left\{ \left| \hat{f}_i^{\text{cal}} - f_i^{\text{cal}} \right| : i = 1, \dots, m \right\} \right),$$

where m is the number of calibration pseudobulks. To allow residual scale to vary with predicted abundance, calibration predictions and test predictions are assigned to the predefined abundance regimes. Each test prediction receives the quantile estimated from its predicted regime; if a calibration regime is empty, all calibration residuals are used as a fallback. The resulting intervals are clipped to $[0, 1]$.

As an uncertainty ablation, we also generate residual-bootstrap intervals from the signed calibration residuals. For each test prediction, 400 residual draws are sampled with replacement, added to the prediction, and clipped through the empirical lower and upper quantiles corresponding to the requested nominal level. These bootstrap intervals are evaluated separately from the conformal intervals and do not alter panel selection, detection labels, or quantification criteria.

4.7 EVALUATION MEASURES

All final metrics are computed on the held-out pseudobulks. Point-estimation performance is summarized by MAE, RMSE, coefficient of determination, Spearman correlation, Pearson correlation, and MAE within low and high fraction ranges. Detection labels are defined by a target fraction of at least 0.35, and detection discrimination is measured by ROC AUC from the predicted fraction. Panel results are compared with the best single-marker predictions using the same held-out pseudobulks.

For each interval level, we report empirical coverage, mean interval width, coverage among fractions at or below 0.50, prediction bias, the false-positive rate among fractions at or below 0.50, and the false-negative rate among fractions at or above the detection threshold. The implementation also records bootstrap confidence intervals for the macro-averaged 80% and 95% coverage values by resampling cluster-level metrics. These quantities are computed after the panel, model, interval procedure, and thresholds have been fixed; held-out outcomes are not used to tune the conformal quantiles or detection threshold.

5 EXPERIMENTAL SETUP

We evaluate the framework on a single-cell PBMC expression dataset stored in H5AD format. Cells are assigned to populations using the `leiden` annotation. These labels define the target

populations, the background populations, and the stratification groups for the discovery-holdout split. The expression matrix is used to derive marker candidates and to simulate bulk-like mixtures; no embedding coordinates are used as model features.

Within each population, 70% of cells are assigned to discovery and 30% to holdout data, with at least two cells retained in each subset. Discovery cells are used for Wilcoxon marker ranking, reference-gene selection, and generation of training and calibration pseudobulks. Held-out cells are used only to generate the final test pseudobulks. Training, calibration, and test pseudobulks are generated independently, with 400, 300, and 500 pseudobulks per target population, respectively, and each pseudobulk contains 500 cells in the primary configuration.

For every target population, we retain at most 30 genes from discovery-only Wilcoxon rankings as marker candidates. We select a reference-gene pool from genes expressed in at least 20% of discovery cells, excluding all candidate markers, and use the three most stable genes for normalization. Candidate panels contain at most five markers. The best single marker is selected using five-fold shuffled cross-validated fraction MAE, while nominal greedy panel expansion requires an MAE improvement of at least 0.002 per added marker. The uncertainty-aware selection objective uses cross-validated MAE, calibration-set MAE for fractions at or below 0.50, 80% and 95% conformal residual widths, and a panel-size penalty of 0.002 per marker, with weights 0.25 on the rare-fraction MAE and 0.10 on the combined interval widths.

Target fractions are sampled from five predefined abundance regimes: absent or trace (0–1%), very low (1–5%), low (5–10%), medium (10–50%), and high (50–90%). The regime probabilities are 0.15, 0.20, 0.20, 0.30, and 0.15, respectively. The sampled fraction is converted to a target-cell count by rounding, after which the remaining cells are allocated among the non-target populations using a Dirichlet draw followed by multinomial sampling. Target and background cells are sampled with replacement from their corresponding discovery or holdout populations.

To model departures from idealized cell averages, the completed robustness configurations apply a nuisance layer whose parameters are fixed before pseudobulk generation. The layer uses log-normal global scaling with standard deviation 0.08, gene-specific multiplicative noise with log-scale standard deviation 0.03, dropout probability 0.02, ambient-RNA fraction 0.01, and a background-mixture Dirichlet concentration of 0.35. The ambient profile is the mean discovery expression profile. Global and gene-specific scaling, dropout, and ambient mixing are applied independently to each generated pseudobulk, while the nuisance parameter values remain fixed across training, calibration, and testing.

For each candidate panel, the mean marker expression minus the mean reference-gene expression is used as a one-dimensional normalized score. A linear regression maps this score to the realized target fraction, and predictions are clipped to $[0, 1]$. The selected panel is fit on the training pseudobulks. Split-conformal intervals use absolute residuals from the independent calibration pseudobulks at nominal levels of 80% and 95%, with quantiles assigned using predefined predicted-abundance strata and clipped to the valid fraction range. The completed uncertainty ablation additionally constructs residual-bootstrap intervals using 400 draws from the signed calibration residuals. Bootstrap resampling with 200 replicates is used for confidence intervals of macro-averaged coverage metrics.

We compare a supplied greedy-panel baseline with uncertainty-aware panel selection under progressively varied configurations. The primary nuisance-robust configuration uses 500 cells per pseudobulk; a finite-cell sensitivity configuration increases the number of cells per pseudobulk and permits zero-target samples; a prediction-stratified configuration assigns calibration and test predictions to the predefined absent-or-trace, very-low, low, medium, and high abundance strata before estimating interval widths; and a final configuration compares residual-bootstrap intervals with the conformal intervals. These configurations preserve discovery-only marker ranking and held-out evaluation while changing the sampling or uncertainty component. The comparisons are intended to distinguish biological marker specificity, point-estimation performance, detection ranking, and interval calibration rather than to identify a single aggregate score as universally optimal.

Final evaluation uses held-out pseudobulks and reports ROC AUC for detection, MAE, RMSE, R^2 , Spearman correlation, Pearson correlation, and MAE in low- and high-fraction subsets. Detection labels use a target fraction threshold of 0.35. Interval evaluation reports empirical coverage, mean interval width, coverage for fractions at or below 0.50, prediction bias, false-positive rate among fractions at or below 0.50, and false-negative rate among fractions at or above the detection threshold. The low- and high-fraction subsets use the configured boundaries of 0.10 and 0.50, respectively,

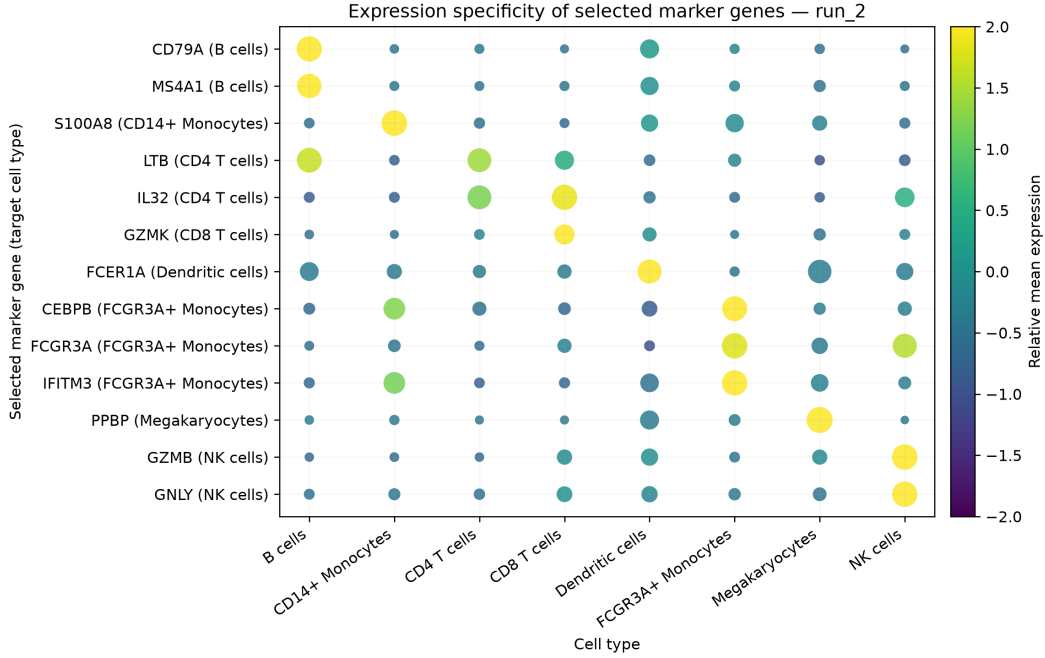


Figure 2: Expression specificity of selected marker genes under fixed assay and background perturbations. Dot size denotes the fraction of expressing cells, and color denotes row-normalized mean expression across cell populations.

while predicted-abundance interval strata use the predefined boundaries 0–0.01, 0.01–0.05, 0.05–0.10, 0.10–0.50, and 0.50–0.90. These thresholds, abundance boundaries, conformal levels, and panel-selection criteria are fixed before held-out metrics are calculated.

6 RESULTS

6.1 MARKER SPECIFICITY AND DETECTION

The selected markers retained target-associated expression patterns across the labeled PBMC populations under the fixed assay and background perturbations. Figure 2 summarizes this behavior: dot size represents the fraction of expressing cells and color represents row-normalized mean expression. The figure supports the biological interpretability of the selected marker-target associations, but also shows why expression specificity alone cannot establish quantitative accuracy or uncertainty calibration.

Under fixed scaling, gene-wise noise, dropout, ambient RNA, and altered background composition, the multi-gene panel achieved a macro ROC AUC of 0.9749, compared with 0.9642 for the best single-marker baseline. The corresponding panel-minus-single-marker AUC difference was 0.0107. Figure 3 shows the per-population ROC curves and confirms that the comparison is heterogeneous across targets rather than a uniform improvement. Detection performance therefore remained strong under perturbation, while the panel advantage was modest in absolute terms.

6.2 QUANTITATIVE FRACTION ESTIMATION

The panel produced a macro MAE of 0.0644, macro RMSE of 0.0946, macro R^2 of 0.8294, macro Spearman correlation of 0.8451, and macro Pearson correlation of 0.9254 under nuisance variation. The corresponding single-marker values were MAE 0.0724, RMSE 0.1061, R^2 0.7839, Spearman correlation 0.8254, and Pearson correlation 0.9038. The panel also reduced high-fraction MAE from 0.1480 to 0.1318 and low-fraction MAE from 0.0496 to 0.0450. Thus, multi-gene aggregation

Held-out cell-type detection performance — run_2

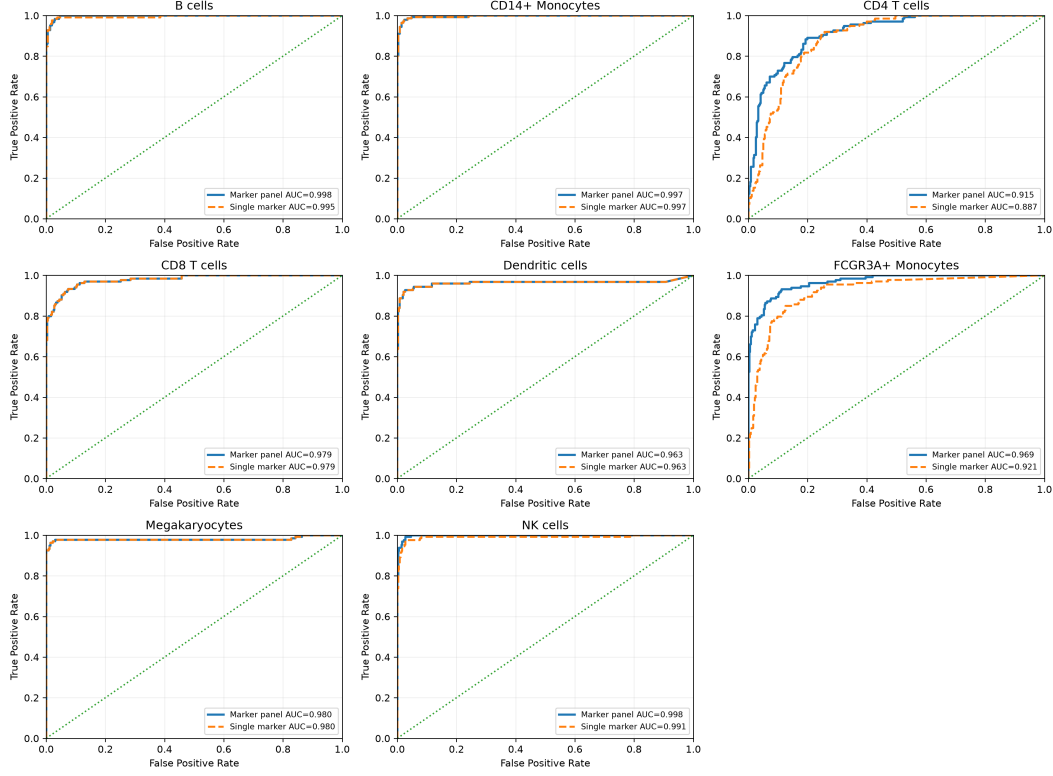


Figure 3: Held-out ROC curves for multi-gene panels and best single-marker baselines under assay and background perturbations. Solid curves denote panels, dashed curves denote single markers, and legends report per-population AUC.

improved the recorded quantitative metrics relative to the single-marker baseline in this configuration, although the residual error remained larger at high abundance than at low abundance.

The true-versus-predicted fractions in Figure 4 show the abundance-dependent structure underlying these summary metrics. The residual plot in Figure 5 makes the same pattern explicit: deviations from zero vary across abundance regimes and target populations. These plots demonstrate that a strong rank correlation or low average MAE does not imply uniform accuracy throughout the concentration range.

6.3 UNCERTAINTY CALIBRATION

The split-conformal intervals did not attain their nominal coverage on held-out pseudobulks. At the 80% nominal level, macro empirical coverage was 70.5% with mean interval width 0.1382; at the 95% nominal level, coverage was 88.8%. The calibration diagnostic in Figure 6 shows both the interval envelopes and coverage by abundance bin. These results indicate that calibration-set residuals transferred imperfectly to held-out samples under the nuisance-perturbed distribution, so nominal conformal levels should not be interpreted as empirically validated coverage in this setting.

Changing the finite-cell configuration altered both point and interval behavior. With the increased-cell and zero-target configuration, macro MAE was 0.0592, macro 80% coverage was 68.9%, macro 95% coverage was 87.4%, and mean 80% interval width was 0.1216. The prediction-stratified calibration configuration left point metrics unchanged but produced 80% and 95% coverage of 66.1% and 86.1%, respectively, with mean 80% width 0.1234. These results do not establish that narrower intervals or higher point accuracy correspond to better calibration; coverage remained below nominal in both configurations.

Quantitative cell-fraction prediction — Conformal intervals (100 cells/pseudobulk)

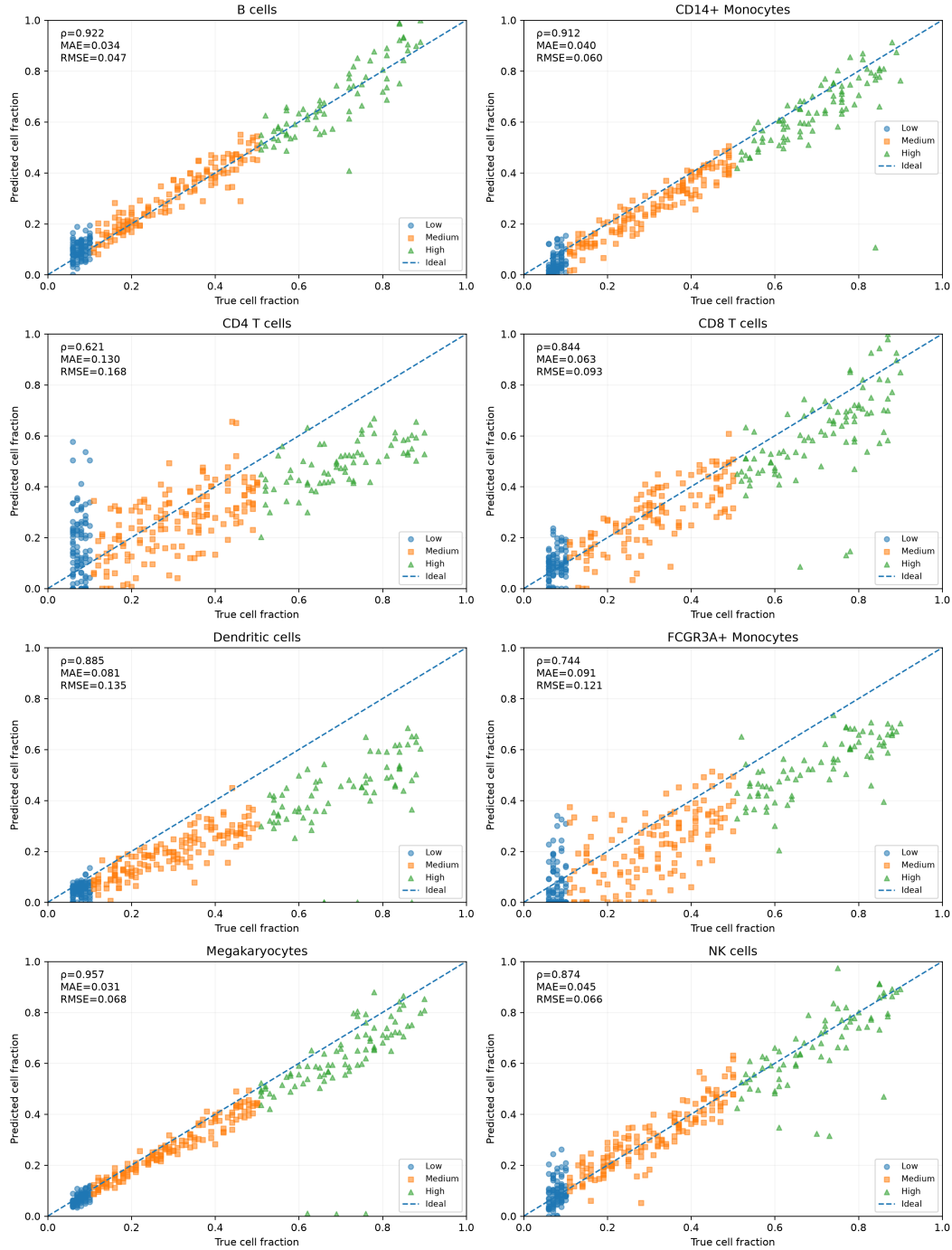


Figure 4: Held-out true versus predicted target fractions under fixed assay and background perturbations. Marker shapes identify abundance regimes, the diagonal denotes ideal prediction, and annotations report per-population Spearman correlation, MAE, and RMSE.

The residual-bootstrap alternative produced macro 80% coverage of 69.1%, macro 95% coverage of 85.0%, and mean 80% interval width 0.1279, compared with 68.5%, 85.9%, and 0.1300 for the corresponding conformal evaluation in the same configuration. The bootstrap therefore gave slightly

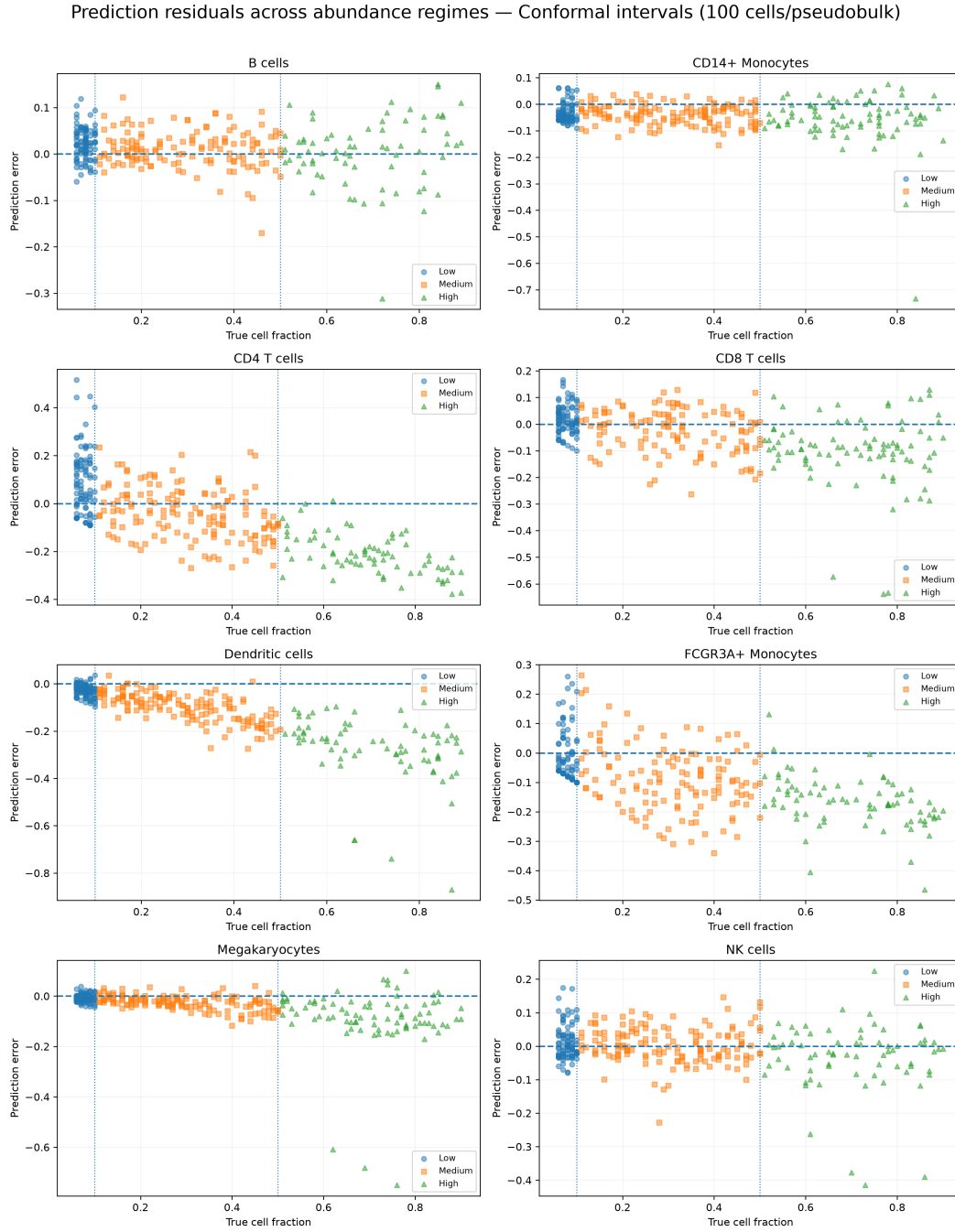


Figure 5: Held-out prediction residuals under fixed assay and background perturbations. Residuals are predicted minus true fraction; the horizontal line denotes zero error and vertical lines mark the configured low- and high-fraction thresholds.

higher 80% coverage and slightly lower 95% coverage, with a modestly narrower mean 80% interval. Because neither procedure reached nominal coverage, the experiment supports treating the bootstrap as a useful comparison rather than as a demonstrated replacement for conformal intervals.

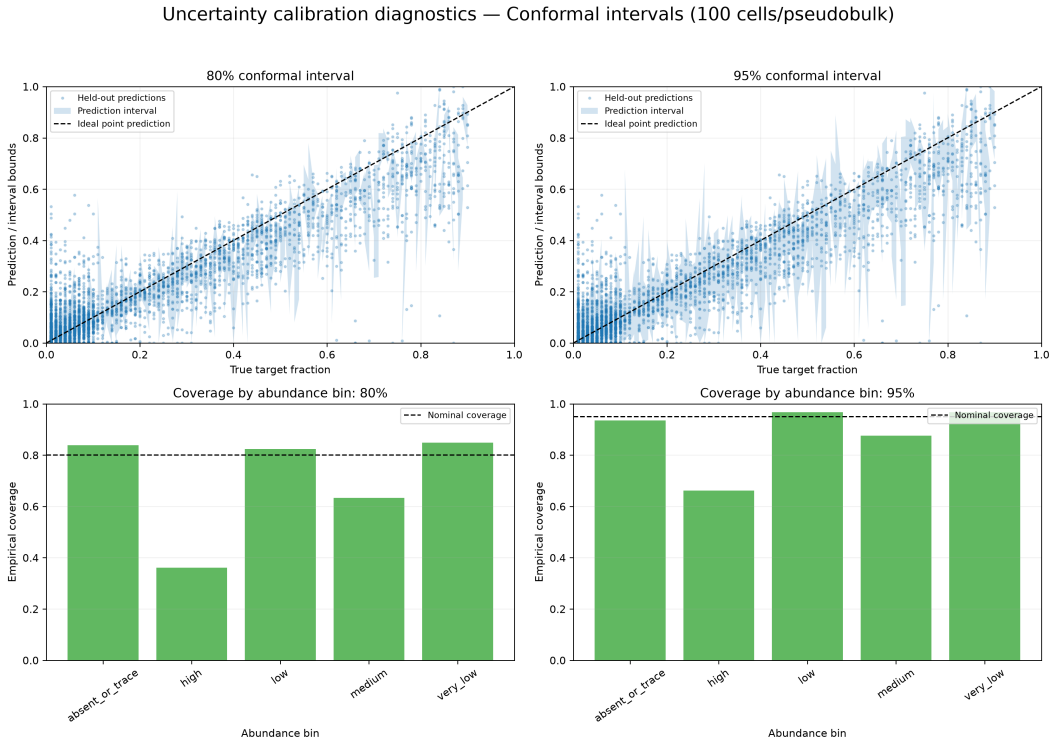


Figure 6: Held-out uncertainty-calibration diagnostics for nominal 80% and 95% intervals. Upper panels show predictions and clipped interval bounds against true fractions; lower panels show empirical coverage by abundance bin with nominal reference lines.

6.4 PANEL COMPARISONS AND CROSS-CONFIGURATION SYNTHESIS

The paired comparison in Figure 8 separates detection from fraction quantification at the population level. Under nuisance variation, the panel improved both AUC and MAE in the macro averages, but the magnitude and direction of the change varied by target. This mixed per-population behavior is consistent with the broader result that panel aggregation can improve quantitative robustness without guaranteeing a large or uniform discrimination gain.

Across the completed configurations, macro panel MAE ranged from 0.0592 to 0.0644, while macro single-marker MAE ranged from 0.0622 to 0.0724. Macro panel-minus-single-marker AUC differences ranged from 0.0008 to 0.0107. *In contrast, macro 80% conformal coverage ranged from 56.2% to 70.5% and macro 95% coverage ranged from 73.8% to 88.8%, all below their nominal levels. Figure 9 summarizes these trade-offs and shows that point accuracy, detection advantage, interval coverage, and interval width respond differently to the*

Although the study defines operational detection and quantification criteria, the completed artifacts report aggregate performance and interval diagnostics rather than a validated sensitivity–specificity curve or a fraction-specific threshold crossing for those criteria. We therefore do not claim a numerical limit of detection or limit of quantification from these experiments. The supported conclusion is narrower: the framework exposes abundance-dependent error and substantial undercoverage, demonstrating why such limits must be estimated explicitly from dedicated, held-out evaluations rather than inferred from point metrics or nominal interval levels.

7 CONCLUSIONS AND FUTURE WORK

We studied the problem of selecting compact molecular marker panels that both discriminate target cell populations and quantify their abundance with explicit uncertainty. Our framework separated discovery-only marker ranking, panel selection, calibration, and held-out evaluation; modeled finite-

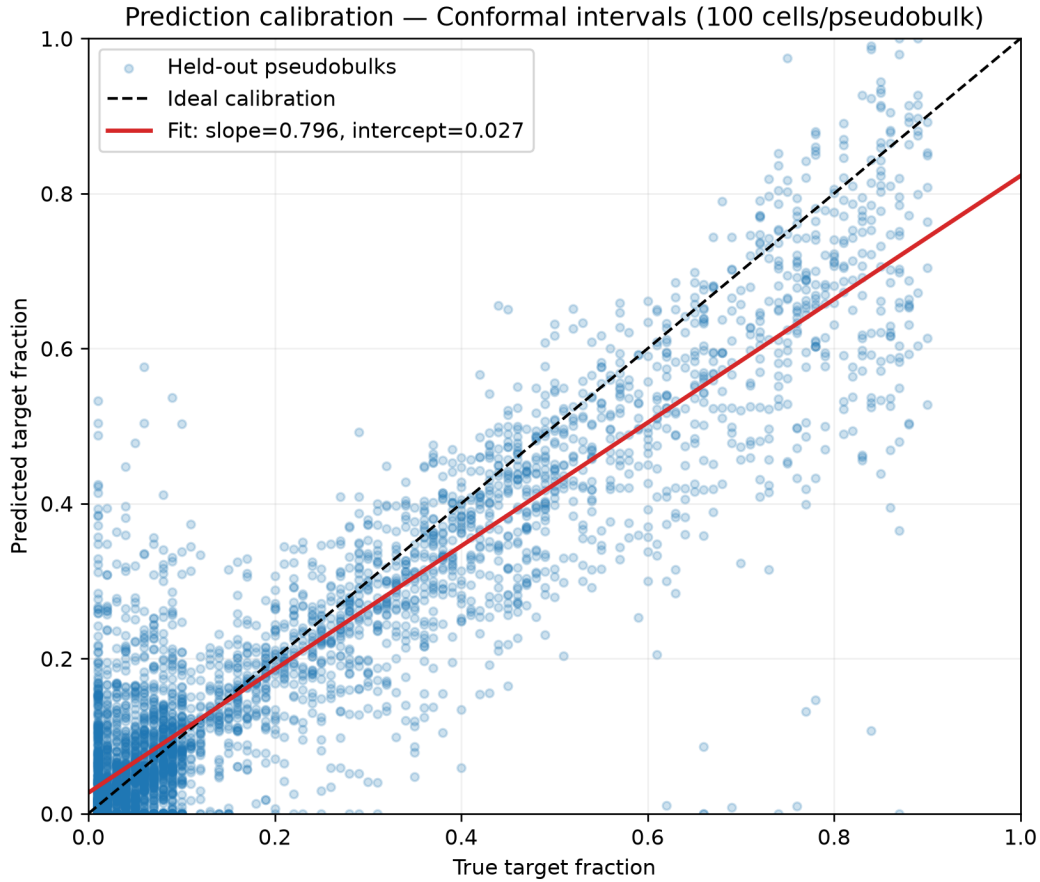


Figure 7: Held-out point-calibration diagram under assay and background perturbations. The identity line denotes ideal calibration, while the fitted line summarizes empirical scaling and offset of predicted fractions.

cell sampling and assay nuisance variation; and compared multi-gene panels with best single-marker baselines using point-estimation, detection, and interval-calibration metrics. This separation was designed to prevent strong discrimination or average error from being mistaken for reliable quantitative inference.

Across the uncertainty-focused configurations (Runs 1–5, excluding the supplied baseline Run 0), multi-gene panels generally improved quantitative point estimates relative to the best single markers, with panel MAE ranging from 0.0592 to 0.0644 and single-marker MAE ranging from 0.0622 to 0.0724. Panel-minus-single-marker AUC differences were positive but modest, ranging from 0.0008 to 0.0107. The central negative result concerned uncertainty: empirical conformal coverage remained below nominal, ranging from 56.2% to 70.5% for nominal 80% intervals and from 73.8% to 88.8% for nominal 95% intervals. The supplied artifacts document generation of residual-bootstrap intervals but do not provide bootstrap coverage results sufficient to assess nominal coverage. These findings show that improved point accuracy and detection ranking do not, by themselves, validate uncertainty estimates.

The experiments provide evidence of abundance-dependent error and undercoverage, but they do not establish a validated numerical limit of detection or limit of quantification. The completed evaluations report aggregate held-out metrics and interval diagnostics rather than a dedicated sensitivity–specificity threshold analysis across sufficiently fine abundance levels. Accordingly, the framework should be viewed as a procedure for estimating and stress-testing empirical assay limits, not as evidence that the particular completed configurations have already achieved validated limits.

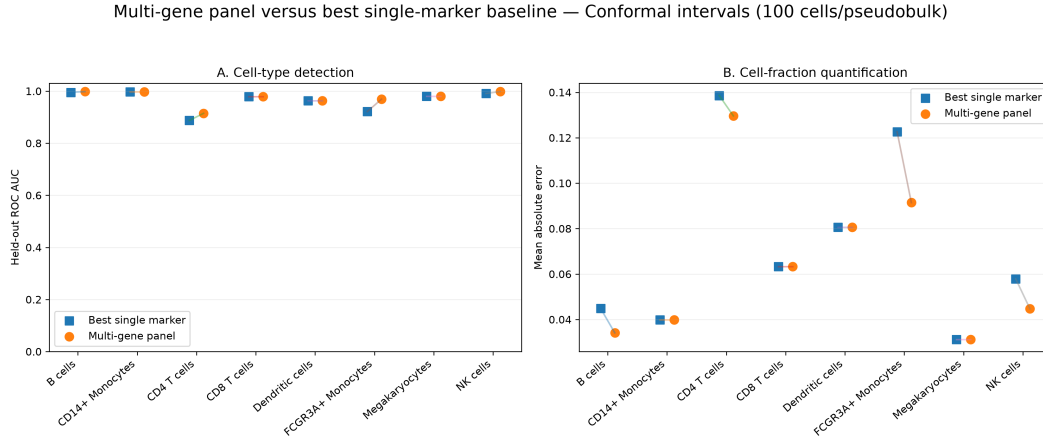


Figure 8: Per-population comparison of multi-gene panels with best single-marker baselines under assay and background perturbations. The left panel reports ROC AUC and the right panel reports MAE; connecting segments show paired changes for each population.

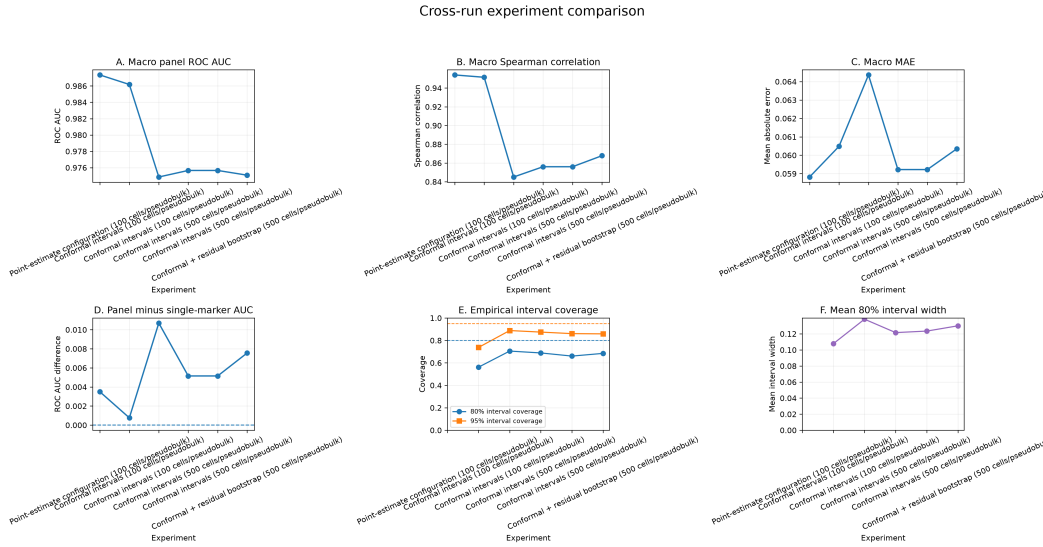


Figure 9: Cross-configuration summary of macro detection, point-estimation, and uncertainty metrics. Coverage panels include nominal reference lines; the figure highlights that improved point metrics or narrower intervals do not by themselves establish calibrated uncertainty.

Future work should extend the evaluation to independent biological datasets, measured qPCR or digital-PCR assays, and repeated nuisance realizations that represent between-run and between-sample variation. Dedicated limit experiments should increase resolution near the detection boundary, prespecify sensitivity, specificity, coverage, and error criteria, and use independent validation samples to estimate confidence intervals for detection and quantification limits. Additional work should also compare conditional calibration methods, assess relative rather than absolute error at very low abundance, and evaluate whether richer models or larger panels improve coverage without producing impractically wide intervals. Such studies would test whether the observed panel benefits and calibration failures persist beyond the simulated PBMC setting.

This work was generated by THE AI SCIENTIST (Lu et al., 2024).

REFERENCES

Chris Lu, Cong Lu, Robert Tjarko Lange, Jakob Foerster, Jeff Clune, and David Ha. The AI Scientist: Towards fully automated open-ended scientific discovery. *arXiv preprint arXiv:2408.06292*, 2024.